

PhishGuard Poster Content Outline

Title: PhishGuard

Subtitle:

Helping users understand suspicious emails, texts, and URLs through explainable phishing risk analysis.

1. Project Overview

PhishGuard is a cybersecurity-focused web application concept designed to help users identify suspicious emails, text messages, and URLs. The project focuses on phishing awareness, explainable risk analysis, and safer digital decision-making.

2. Problem

Phishing attacks often use suspicious links, fake messages, impersonation, urgent language, and requests for personal information. Many users may recognize that something looks suspicious, but they may not understand why it is risky or what action they should take next.

3. Project Goal

PhishGuard aims to help users:

Recognize common phishing indicators.

Understand why a message or URL may be suspicious.

Receive clear safety guidance before clicking links or sharing information.

Learn phishing warning signs through simple explanations.

4. Prototype Focus

The current prototype focuses on suspicious URL analysis. Users can enter a URL, receive a risk result, and view warning signs connected to the scan.

Main prototype areas:

URL input

URL scan result

Risk score or risk level

Sonar Analysis Log

Recommended user actions

Educational phishing tips

5. Key Feature: Sonar Analysis Log

The Sonar Analysis Log displays warning signs found during a scan. These warnings may include missing HTTPS, suspicious keywords, long URLs, URL shorteners, redirects, or other suspicious link behavior.

Purpose:

To show users what was detected instead of only giving a general risk result.

6. Proposed Feature: Bait Breakdown

Bait Breakdown is a proposed feature that expands the Sonar Analysis Log by explaining each warning in simple language.

Each warning would explain:

What was detected

Why it may be risky

What action the user should take next

Example:

Warning:

URL Shortener Detected

Explanation:

Shortened links can hide the real destination of a website.

Recommended Action:

Do not open the link unless the sender or destination can be verified.

7. Research Support

The research supports PhishGuard by showing the importance of phishing awareness, clear warning explanations, risk scoring, safe URL handling, and real-world phishing examples.

Research areas used:

Phishing warning signs

User safety guidance

SMS phishing and URL-based attacks

Explainable risk analysis

Phishing training and awareness

Risk scoring methods

Secure URL handling

Real-world phishing case studies

8. Agile/Scrum Process

The team used Agile/Scrum methods to organize and track project progress.

Sprint work included:

Sprint planning

Backlog grooming

Daily scrum updates

Sprint review

User stories

Acceptance criteria

Documentation and artifacts

Prototype planning

QA planning

9. Team Contributions

Product Owner:

Guided project priorities and user story selection.

Team Lead/Scrum Coordinator:

Helped manage sprint progress, deadlines, and required artifacts.

Research and Documentation:

Researched phishing risks, organized sources, created documentation, and supported feature descriptions.

Design/UX:

Worked on wireframes, layout ideas, and user interface direction.

QA/Requirements:

Supported user stories, test planning, and acceptance criteria.

10. Future Work

Continue developing the website prototype.

Improve URL testing methods.

Expand scan result explanations.

Refine the Bait Breakdown feature.

Improve educational resources and user guidance.

Continue QA testing and prototype feedback.

11. Conclusion

PhishGuard supports safer digital decision-making by helping users recognize suspicious content and understand why it may be risky. The project combines phishing awareness, explainable scan results, and user guidance while staying realistic for the Capstone 1 planning and prototype phase.